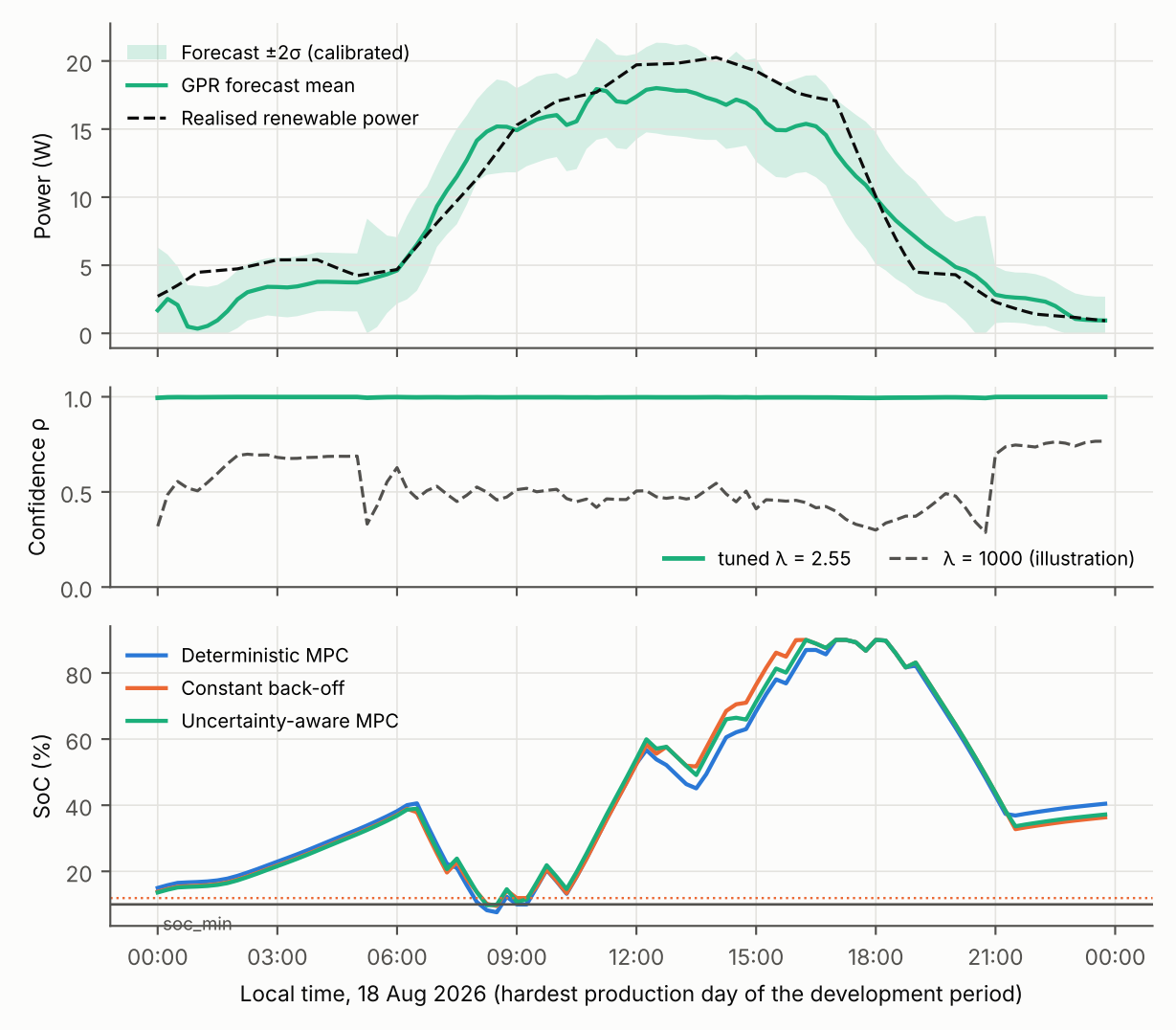
Forecast ±2σ (calibrated)
GPR forecast mean
Realised renewable power
Power (W)
0
5
10
15
20
Confidence ρ
0.0
0.5
1.0
tuned λ = 2.55
λ = 1000 (illustration)
Deterministic MPC
Constant back-off
Uncertainty-aware MPC
SoC (%)
20
40
60
80
soc_min
00:00
03:00
06:00
09:00
12:00
15:00
18:00
21:00
00:00
Local time, 18 Aug 2026 (hardest production day of the development period)